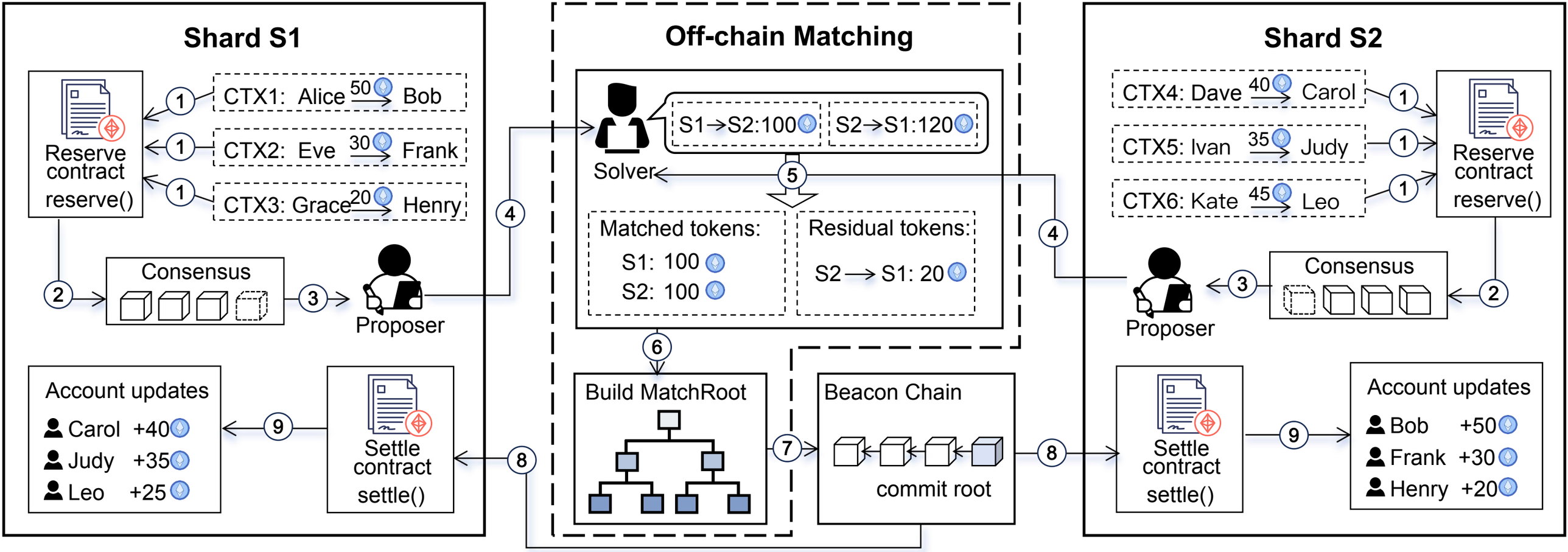

Shard S1
Reserve contract
reserve()
CTX1: Alice 50 Bob
CTX2: Eve 30 Frank
CTX3: Grace 20 Henry
Consensus
Proposer
Account updates
Carol +40
Judy +35
Leo +25
Settle contract
settle()
Off-chain Matching
Solver
S1→S2:100
S2→S1:120
Matched tokens:
S1: 100
S2: 100
Residual tokens:
S2 → S1: 20
Build MatchRoot
Beacon Chain
commit root
Shard S2
CTX4: Dave 40 Carol
CTX5: Ivan 35 Judy
CTX6: Kate 45 Leo
Reserve contract
reserve()
Consensus
Proposer
Settle contract
settle()
Account updates
Bob +50
Frank +30
Henry +20